

Initial Errors running on Ubuntu

```
bkatsu@bensurface:~/MA1$ cmake --build build
[ 50%] Building CXX object CMakeFiles/MA1.dir/main.cpp.o
/home/bkatsu/MA1/main.cpp:12:1: error: new types may not be defined in a return type
12 | class queue
    | ^~~~~~
/home/bkatsu/MA1/main.cpp:12:1: note: (perhaps a semicolon is missing after the definition of 'queue')
/home/bkatsu/MA1/main.cpp:12:1: error: return type specification for constructor invalid
/home/bkatsu/MA1/main.cpp: In member function 'void queue::dequeue()':
/home/bkatsu/MA1/main.cpp:59:24: error: invalid operands of types 'const char [10]' and 'int' to binary 'operator<<'
59 |     cout < "Removing " << arr[front] << '\n';
    |                      ^~
    |                      |
    |          const char [10]      int
/home/bkatsu/MA1/main.cpp: In member function 'void queue::enqueue(int)':
/home/bkatsu/MA1/main.cpp:69:9: error: 'isFul' was not declared in this scope; did you mean 'isFull'?
69 |     if (isFul())
    |         ^~~~~
    |         isFull
/home/bkatsu/MA1/main.cpp: In member function 'int queue::peek()':
/home/bkatsu/MA1/main.cpp:88:16: error: 'numeric_limits' was not declared in this scope
88 |     return numeric_limits<int>::min();
    |            ^~~~~~
/home/bkatsu/MA1/main.cpp:88:31: error: expected primary-expression before 'int'
88 |     return numeric_limits<int>::min();
    |                               ^~~~
/home/bkatsu/MA1/main.cpp:88:31: error: expected ';' before 'int'
88 |     return numeric_limits<int>::min();
    |                               ^~~~
    |                               ;
/home/bkatsu/MA1/main.cpp:88:34: error: expected unqualified-id before '>' token
88 |     return numeric_limits<int>::min();
    |                               ^
/home/bkatsu/MA1/main.cpp: In member function 'bool queue::isFull()':
/home/bkatsu/MA1/main.cpp:108:19: error: lvalue required as left operand of assignment
108 |     return (size()-1 = capacity);
    |                   ^~
gmake[2]: *** [CMakeFiles/MA1.dir/build.make:76: CMakeFiles/MA1.dir/main.cpp.o] Error 1
gmake[1]: *** [CMakeFiles/Makefile2:83: CMakeFiles/MA1.dir/all] Error 2
gmake: *** [Makefile:91: all] Error 2
bkatsu@bensurface:~/MA1$
```

Debugging using GDB on the UNIX terminal. Debugging testQueue::testenqueue()

```
bkatsu@bensurface: ~/MA1
Inserting 3
Inserting 2
Inserting 1
OverFlow
Program Terminated

Breakpoint 1, testQueue::testenqueue (this=0x7fffffffef1b7) at /home/bkatsu/MA1/testQueue.cpp:178
178 {
(gdb) next
179     cout <<"testenqueue()" <<endl;
(gdb) next
testenqueue()
180     queue test(3);
(gdb) next
181     test.enqueue(2);
(gdb) next
Inserting 2
182     test.enqueue(1);
(gdb) next
Inserting 1
183 }
(gdb) p test
$1 = {arr = 0x5555555555b2c0, capacity = 3, front = 0, rear = 1, count = 2}
(gdb) p test.size()
$2 = 2
(gdb) p test.isFull()
$3 = false
(gdb) p test.peek()
$4 = 1
(gdb) |
```